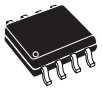
MC1403

2.5V PRECISION SERIAL VOLTAGE REFERENCE

- OUTPUT VOLTAGE : 2.5V ±1%
- INPUT VOLTAGE RANGE : 4.5V to 40V
- QUIESCENT CURRENT : 1.2mA typ.
- OUTPUT CURRENT : 10mA



D
SO8
(Plastic Micropackage)

DESCRIPTION

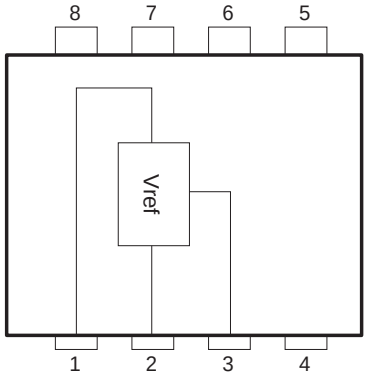
The MC1403 is a serial 2.5V bandgap voltage reference. The major advantages are the 1% precision, the wide input voltage range (4.5V to 40V) and the low quiescent current (1.5mA max.)

ORDER CODE

Part Number	Temperature Range	Package
		D
MC1403	0°C, +70°C	•

D = Small Outline Package (SO) - also available in Tape & Reel (DT)

PIN CONNECTIONS (top view)



1 - V_{in}
2 - V_{out}
3 - GND
4 - N.C.
5 - N.C.
6 - N.C.
7 - N.C.
8 - N.C.

ABSOLUTE MAXIMUM RATINGS

Symbol	Parameter	Value	Unit
V_{IN}	Input Voltage	40	V
T_{stg}	Storage Temperature Range	-65 to +150	°C

OPERATING CONDITIONS

Symbol	Parameter	Value	Unit
V_{IN}	Input Voltage (I_{OUT} =0mA)	4.5 to 40	V
T_{oper}	Operating Free Air Temperature Range	0 to +70	°C

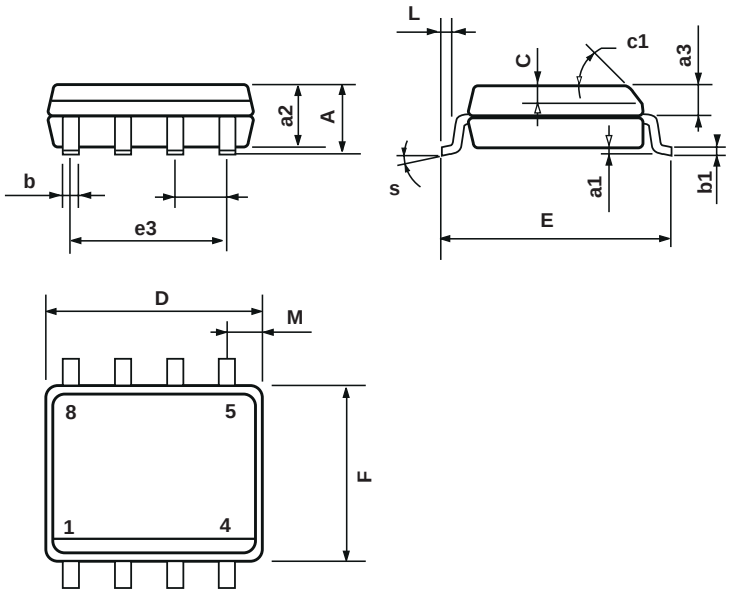
ELECTRICAL CHARACTERISTICS

T_{amb} = 25°C, V_{IN} =15V (unless otherwise specified)

Symbol	Parameter	Min.	Typ.	Max.	Unit
V_{OUT}	Output Voltage I_{OUT} = 0mA	2.475	2.5	2.525	V
$\frac{\Delta Vout}{\Delta T}$	Temperature Coefficient of Output Voltage I_{OUT} = 0mA		10	40	ppm/°C
ΔV_{OUT}	Output Voltage change Over the Full Temperature Range I_{OUT} = 0mA		2	7	mV
Reg_{line}	Line regulation I_{OUT} = 0mA, 4.5V ≤V_i ≤15V I_{OUT} = 0mA, 15V ≤V_i ≤40V			3 4.5	mV
Reg_{load}	Load Regulation 0 ≤ I_{OUT} ≤10mA			10	mV
I_Q	Quiescent Current I_{OUT} = 0mA		1.2	1.5	mA

PACKAGE MECHANICAL DATA

8 PINS - PLASTIC MICROPACKAGE (SO)



Dimensions	Millimeters			Inches		
	Min.	Typ.	Max.	Min.	Typ.	Max.
A			1.75			0.069
a1	0.1		0.25	0.004		0.010
a2			1.65			0.065
a3	0.65		0.85	0.026		0.033
b	0.35		0.48	0.014		0.019
b1	0.19		0.25	0.007		0.010
C	0.25		0.5	0.010		0.020
c1	45° (typ.)					
D	4.8		5.0	0.189		0.197
E	5.8		6.2	0.228		0.244
e		1.27			0.050	
e3		3.81			0.150	
F	3.8		4.0	0.150		0.157
L	0.4		1.27	0.016		0.050
M			0.6			0.024
S	8° (max.)					

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